

Testing a simple calculator web application involves ensuring that all functionalities work correctly, that the UI is intuitive and responsive, and that the application is accessible. Below are some test cases to cover the application thoroughly:

Functional Test Cases

1. Basic Arithmetic Operations **Checked and worked. 6.6.2024**

- **Addition:** Verify that the addition operation returns the correct result (e.g., $2 + 3 = 5$).
- **Subtraction:** Verify that the subtraction operation returns the correct result (e.g., $5 - 3 = 2$).
- **Multiplication:** Verify that the multiplication operation returns the correct result (e.g., $4 * 3 = 12$).
- **Division:** Verify that the division operation returns the correct result (e.g., $12 / 4 = 3$).

2. Order of Operations **Checked and worked. 6.6.2024**

- Verify that the calculator respects the order of operations (e.g., $2 + 3 * 4 = 14$).

3. Edge Cases **Checked and worked. 6.6.2024**

- **Division by Zero:** Verify that the calculator handles division by zero appropriately (e.g., $5 / 0$ should display an error or a specific message).
- **Multiple Operations:** Verify that the calculator can handle multiple operations sequentially (e.g., $2 + 3 - 1 * 4 / 2$).

4. Display **Checked and worked. 6.6.2024**

- Verify that the display updates correctly with each button press.
- Verify that the display clears correctly when the 'C' button is pressed.

5. Keyboard Support **Checked and worked. 6.6.2024**

- Verify that all buttons can be operated via keyboard inputs.
- Verify that the 'Escape' key clears the display.

Usability Test Cases **Checked and worked. 6.6.2024**

1. Button Size and Spacing

- Verify that all buttons are of the same size and are evenly spaced.

2. Theme Toggle **Checked and worked. 6.6.2024**

- Verify that the theme toggles between light and dark correctly.
- Verify that the button label changes appropriately.

3. Error Handling **Checked and worked. 6.6.2024**

- Verify that the calculator displays a user-friendly message for invalid operations (e.g., $2 + + 3$).

Accessibility Test Cases

1. **Screen Reader Compatibility** **Checked and worked. 6.6.2024**
 - Verify that all buttons are labelled correctly for screen readers.
 - Verify that the display updates are announced by the screen reader.
2. **Keyboard Navigation** **Checked and worked. 6.6.2024**
 - Verify that the calculator can be fully operated using only the keyboard.
 - Verify that focus states are visible and clear.

Cross-Browser Compatibility Test Cases

1. **Different Browsers** **Checked and worked. 6.6.2024**
 - Verify that the calculator works correctly on different browsers (e.g., Chrome and firefox)
2. **Different Devices** **Checked and worked. 6.6.2024**
 - Verify that the calculator is responsive and works correctly on different devices (e.g., desktops, tablets, smartphones).

Performance Test Cases

1. **Load Time** **Checked and worked. 6.6.2024**
 - Verify that the calculator loads quickly and is responsive to user inputs.
2. **Stress Testing** **Checked and worked. 6.6.2024**
 - Verify that the calculator can handle rapid inputs without crashing or slowing down.

Example of Specific Test Cases

1. **Test Case: Addition Operation** **Checked and worked. 6.6.2024**
 - **Precondition:** Calculator is loaded.
 - **Steps:**
 1. Press '2'.
 2. Press '+'
 3. Press '3'.
 4. Press '='.
 - **Expected Result:** Display shows '5'.
2. **Test Case: Division by Zero** **Checked and worked. 6.6.2024**
 - **Precondition:** Calculator is loaded.
 - **Steps:**

1. Press '5'.
2. Press '/'.
3. Press '0'.
4. Press '='.

- **Expected Result:** Display shows an error message or 'Infinity'.

3. Test Case: Toggle Theme **Checked and worked. 6.6.2024**

- **Precondition:** Calculator is loaded.
- **Steps:**
 1. Press the 'Dark / Light' button.
- **Expected Result:** Theme switches to dark mode.
- **Steps:** 2. Press the 'Dark / Light' button again.
- **Expected Result:** Theme switches to light mode.

4. Test Case: Screen Reader Compatibility **Checked and worked. 6.6.2024**

- **Precondition:** The calculator is loaded, and the screen reader is enabled.
- **Steps:**
 1. Use the screen reader to navigate to the '2' button.
 2. Press 'Enter'.
 3. Use the screen reader to navigate to the '+' button.
 4. Press 'Enter'.
 5. Use the screen reader to navigate to the '3' button.
 6. Press 'Enter'.
 7. Use the screen reader to navigate to the '=' button.
 8. Press 'Enter'.
- **Expected Result:** Screen reader announces '5'. **Checked and worked. 6.6.2024**